



AI behind 'deepfakes' exploited further

The AI tech behind scary-real celebrity 'deepfakes' is being used to create completely fictitious faces, cats, and Airbnb listings <https://digit.in/mar19-38>



Amazon reveals energy plan

After being criticized by Greenpeace Amazon plans to cut carbon emissions in half by 2030. <https://digit.in/mar19-39>

shots. At the best possible composition, the white line and the dot turns yellow.

Professional photographers use different settings depending on their subjects. Landscapes, portraits or a boat in the ocean may require adjustments to white balance, saturation and brightness to get the best possible photo. Not all smartphone users are familiar with these settings, which is why Samsung uses a feature called Scene Optimiser. Samsung used information from surveys of the most appropriate colour and tone adjustments for different kinds of photos, and fed it into the machine learning algorithm. When the photo is being taken, the phone automatically understands the content of the images, and adjusts the settings automatically. The S10 supports 30 modes for a range of subjects. Dogs and cats, sunrises and sunsets, trees and flowers, all have their own settings.

Honor devices use a similar approach to adjust contrast, brightness and saturation. The View 20 supports 60 different scenarios for optimisation. Honor uses a slightly different approach than Samsung, by tagging the scene with attributes, such as sunset and clouds, to support what it calls Multiple Scene Recognition. Additionally, Honor devices can recognise and separate the individual elements of an image, and apply tweaks to each of these layers. If a scene contains mountains, trees, clouds, water and a building, it then optimises each of the elements individually, in real time. The process of isolating or masking each individual element is called Semantic Image Segmentation.

Another AI feature on Galaxy devices is the flaw detection feature, that informs the user that the captured photo may not be of the best quality. This is when a person may have

blinked, the photo is blurry, or part of the finger is covering the lens. An alert shows up on screen as soon as the image is clicked.

The Pixel 3 uses AI for a feature known as the Top Shot. The camera saves images continuously from 1.5 seconds before to 1.5 seconds after the button is pressed. Up to 90 images may be captured during this period. AI automatically chooses two additional frames, apart from the one clicked, as suggestions for better photographs. If the subject is a person, the AI checks if the eyes are open and if the person is smiling. The AI checks the pixels across these shots to determine if the subject is moving, and uses information from the gyroscope to check if the camera itself is moving. Finally, there is a "3A" score for the auto white balance, autofocus and auto exposure. All these factors are considered before two alternative frames are selected. **4**

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